Junseo Lee

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Room 203, Bldg. 138, Institute of Computer Technology,
Seoul National University, Seoul 08826, Korea

Education

Harvard University, Harvard Kenneth C. Griffin Graduate School of Arts and Sciences — Cambridge, MA
Doctor of Philosophy (Ph.D.), Quantum Science and Engineering — Starting Fall 2026

- Advisors: Anurag Anshu and Sitan Chen (Computer Science), Jordan Cotler (Theoretical Physics)
- Affiliated with the Harvard Quantum Initiative and the Theory of Computation Group

Yonsei University, College of Engineering — Seoul, Korea
Bachelor of Science (B.S.), Electrical and Electronic Engineering — Mar. 2019 – Feb. 2023

- Fully funded by the Hyundai Motor CMK Science and Technology Scholarship
- Thesis: *Combinatorial Designs in Information Theory*

Chungnam Science High School — Gongju, Korea
Concentration in Mathematics, Early Honors Graduate — Mar. 2017 – Dec. 2018

Research Interests

Quantum Complexity Theory, Quantum Learning Theory, Quantum Many-Body Physics, Quantum Algorithms

Research Experience and Positions

Seoul National University (SNU) — Seoul, Korea
Research Fellow, Institute of Computer Technology (Host: Kabgyun Jeong) — Apr. 2026 – Present

- Funded by the National Research Foundation of Korea (NRF), the Institute for Information and Communications Technology Planning and Evaluation (IITP), and the Electronics and Telecommunications Research Institute (ETRI)

Research Affiliate, Research Institute of Mathematics — Jan. 2023 – Mar. 2026
Research Assistant, Department of Mathematical Sciences — Mar. 2020 – Dec. 2022

Republic of Korea Army — Seoul, Korea
Research Scientist (Technical Research Personnel) — Mar. 2023 – Mar. 2026

- Mandatory military service fulfilled through a government-designated research program for scientists

Yonsei University — Seoul, Korea
Research Assistant, School of Electrical and Electronic Engineering (Advisor: Chulhee Lee) — Jul. 2022 – Dec. 2022
Research Assistant, School of Mathematics and Computing (Advisor: Jeehyun Lee) — Dec. 2021 – Feb. 2022

Publications (Google Scholar Profile)

*Equal contribution. †Authors listed alphabetically.

Preprints

(17) Myeongjin Shin, **Junseo Lee**, and Changhun Oh, "Learning Hamiltonians in the Heisenberg limit with long-time dynamics," (2026). In preparation.

(16) Kabgyun Jeong, Donghwa Ji, **Junseo Lee**†, Adam Sawicki, and Oskar Słowik, "Explicit bounds on polylogarithmic spectral gap decay in unitary channels," (2026). In preparation.

(15) Andreas Bluhm, Matthias C. Caro, Francisco Escudero Gutiérrez, **Junseo Lee**†, Aadil Oufkir, Cambyse Rouzé, and Myeongjin Shin, "Certifying and learning local quantum Hamiltonians," (2026). arXiv:2603.29809.
► Accepted for a contributed talk at **TQC 2026**.
► This work subsumes a previous note (joint with M. Shin), arXiv:2512.09778.

(14) Marco Fanizza, Vishnu Iyer, **Junseo Lee**†, Antonio Anna Mele, and Francesco Anna Mele, "Efficient learning of bosonic Gaussian unitaries," (2025). arXiv:2510.05531.
► Accepted for a contributed talk at **QIP 2026**.

(13) Kartik Anand, Kabgyun Jeong, and **Junseo Lee**†, "Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy," (2025). arXiv:2506.19792.

Journal Articles

(12) Donghwa Ji, **Junseo Lee**, Myeongjin Shin, IlKwon Sohn, and Kabgyun Jeong, "Bounding quantum uncommon information with quantum neural estimators," *Quantum Science and Technology* **11**, 015001 (2026). DOI: 10.1088/2058-9565/ae18f4.

(11) Myeongjin Shin*, **Junseo Lee***, Seungwoo Lee, and Kabgyun Jeong, "Resource-efficient algorithm for estimating the trace of quantum state powers," *Quantum* **9**, 1832 (2025). DOI: 10.22331/q-2025-08-27-1832.

(10) Mingyu Lee, Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, "Mutual information maximizing quantum generative adversarial networks," *Scientific Reports* **15**, 32835 (2025). DOI: 10.1038/s41598-025-18476-y.

(9) Myeongjin Shin*, Seungwoo Lee*, **Junseo Lee***, Donghwa Ji, Hyeonjun Yeo, and Kabgyun Jeong, "Disentanglement provides a unified estimation for quantum entropies and distance measures," *Physical Review A* **110**, 062418 (2024). DOI: 10.1103/PhysRevA.110.062418.

(8) Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, "Estimating quantum mutual information through a quantum neural network," *Quantum Information Processing* **23**, 57 (2024). DOI: 10.1007/s11128-023-04253-1.

(7) **Junseo Lee** and Kabgyun Jeong, "Quantum Rényi entropy functionals for bosonic gaussian systems," *Physics Letters A* **490**, 129183 (2023). DOI: 10.1016/j.physleta.2023.129183
► PLA Special Issue, Foundations and applications of Quantum Optics (2024).

(6) **Junseo Lee**, Hyeonjun Yeo, and Kabgyun Jeong, "Weighted p-Rényi entropy power inequality: Information theory to quantum Shannon theory," *International Journal of Theoretical Physics* **62**, 253 (2023). DOI: 10.1007/s10773-023-05512-8

(5) **Junseo Lee** and Kabgyun Jeong, "High-dimensional private quantum channels and regular polytopes," *Communications in Physics* **31**, 189 (2021). DOI: 10.15625/0868-3166/15762
► Third Prize, Undergraduate Research Exhibition, Korean Physical Society (2021).

(4) Kabgyun Jeong, **Junseo Lee**, *et al.*, "Single qubit private quantum channels and 3-dimensional regular polyhedra," *New Physics: Sae Mulli* **68**, 232 (2018). DOI: 10.3938/NPSM.68.232
► Bronze Award, The Humantech Paper Award, Samsung Electronics (2018).

Book Chapters

(3) **Junseo Lee**, "Assessing Quantum Integer Factorization Performance with Shor's Algorithm," In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*, Edited by Mohammad Hammoudeh, Abdullah T. Alessa, Amro M. Sherbeeni, Clinton M. Firth, Abdullah S. Alessa, *CRC Press* (2024). DOI: 10.1201/9781003475286

Patents

(2) Kabgyun Jeong, Myeongjin Shin, and **Junseo Lee**, "Method for estimating quantum mutual information through a quantum neural network," *Korea Patent* Open No. 10-2026-0009068 (2024). DOI: 10.8080/1020240091151

Notes

(1) **Junseo Lee**† and Myeongjin Shin, "Optimal certification of constant-local Hamiltonians," (2025). arXiv:2512.09778.
► Best Presentation Award, Annual Meeting of the Quantum Information Society of Korea (QISK 2026).
► This note has been subsumed by a subsequent work. See the published version at arXiv:2603.29809.

Professional Activities

Journal Reviewer

- **American Physical Society:** Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A
- **Nature Research:** npj Quantum Information
- **Institute of Electrical and Electronics Engineers:** IEEE Transactions on Information Theory
- **Others:** Quantum, Physics Letters A, Annalen der Physik

Conference Reviewer

- Theory of Quantum Computation, Communication and Cryptography (TQC), 2026
- Quantum Computing Theory in Practice (QCTiP), 2026
- Quantum Techniques in Machine Learning (QTML), 2025

Organizing

- *Organizer*, QISCA Winter School on Quantum Learning Theory for Bosonic and Fermionic Systems, 2026
- *Co-organizer*, SNU Quantum Information Theory Seminar, 2024–2025
- *Co-organizer*, Quantum AI Hackathon, jointly organized by Kakao Enterprise Corp. and Jeonju University, 2025

Committee and Reviewing

- *Poster Session Judge*, National Undergraduate Quantum Conference, Seoul National University, 2026
- *Selection Committee*, Quantum Internship Program, National Information Society Agency and Korea Quantum Industry Center, 2024–2025

Outreach and Community Resources

- *Creator and Maintainer*, Quantum Learning Theory Zoo, 2025–
- *Facilitator (Mentor)*, Mathematics Section, Korea Scholar's Conference for Youth (KSCY), Yonsei University, 2019

Selected Honors and Awards

Funding and Fellowships

- *Harvard Graduate Fellowship*, Harvard University, 2026
- *Academic Travel Grant*, Hyundai Motor CMK Foundation, 2022
- *Hyundai Motor CMK Science and Technology Scholarship*, Hyundai Motor CMK Foundation, 2021–2022
- *Teaching Fellowship for Software Courses*, Yonsei University, 2021–2022

Academic Excellence Awards

- *High Honor Student*, Yonsei University, 2022
- *Honor Student*, Yonsei University, 2020–2021
- *JH Science Scholarship*, JH Foundation, 2017
- *Academic Excellence Scholarship*, Asan Future Scholarship Foundation, 2016

Additional Honors and Awards

- *Selected Paper Award*, Finance and Economics Contest, DB Group, 2022
- *Best Tutor Award*, Yonsei University, 2021–2022
- *Third Prize*, Undergraduate Research Exhibition, Korean Physical Society, 2021
- *Bronze Award*, The Humantech Paper Award, Samsung Electronics, 2018
- *Best Translator Award*, NAVER Connect Foundation and Khan Academy, 2018
- *National Honorable Mention and Regional Gold Award*, Korean Olympiad in Informatics, 2016

Research Visits

- Harvard University, hosted by Anurag Anshu and the Harvard Quantum Initiative, Mar. 2026

Teaching Experience

Undergraduate Research Mentoring

(Mentored undergraduate students during my appointments as Research Affiliate and Research Fellow at SNU)

- Kartik Anand (IIT Goa CS), research in quantum complexity theory [arXiv '25], 2025–present
- Donghwa Ji (SNU Math), research in quantum information theory [Quantum Sci. Technol. '26], 2025–present (co-supervised with Kabgyun Jeong)
- Mingyu Lee (SNU CS), research in quantum machine learning and algorithms [Sci. Rep. '25], 2024–present (Research Intern at University of Oxford; co-supervised with Sathyawageeswar Subramanian)
- Myeongjin Shin (KAIST CS), research in quantum learning theory [PRA '24, Quantum '25, TQC '26], 2023–present (co-supervised with Kabgyun Jeong)

Quantum Information Science Club Association

(Teaching materials are available at: harris-junseo-lee.github.io/teaching/)

- *Organizer and Lecturer*, Quantum Learning Theory for Bosonic and Fermionic Systems, Lecture 6: "Learning Bosonic Gaussian Unitaries", Winter 2026
- *Invited Lecturer*, Quantum Complexity Reading Group, Fall 2025
- *Invited Lecturer*, Quantum Learning and Complexity Theory, Summer 2025

University–Industry Research Internship

- *Instructor*, AAA558/AAA559: College of Informatics Internship, Korea University (Graduate Course), Fall 2025
- *Instructor*, SW4343: Software Field Placement 1, Korea Aerospace University, Fall 2024

Yonsei University

- *Teaching Assistant*, YCS1009: Change the World through Programming, Fall 2022
- *Teaching Assistant*, YCS1002: Software Programming, Fall 2022
- *Teaching Assistant*, EEE1108: Engineering Information Processing, Fall 2021
- *Course Tutor*, MAT2016: Engineering Mathematics 3, Spring 2022, Best Tutor Award
- *Course Tutor*, MAT1012: Engineering Mathematics 2, Fall 2021, Best Tutor Award

Selected Talks

*Online talk.

Research Talks

"Certifying and learning local quantum Hamiltonians"

- *Contributed talk*, **TQC 2026**, Aug. 2026

"Optimal certification of constant-local Hamiltonians"

- *Invited talk*, Quantum Software Lab Seminar, University of Edinburgh, Mar. 2026*
- *Contributed talk*, Annual Meeting of the Quantum Information Society of Korea, Feb. 2026

"Efficient learning of bosonic Gaussian unitaries"

- *Invited talk*, Annual Meeting of the Quantum Information Society of Korea, Feb. 2026
- *Invited talk*, N^3etFraST Workshop, Nov. 2025
- *Invited talk*, Yonsei Quantum Data Science & AI Lab Seminar, Nov. 2025
- *Contributed talk*, **QIP 2026**, Jan. 2026 (Presented under the title "Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators", as a merged submission with arXiv:2504.11318.)

"Resource-efficient algorithm for estimating the trace of quantum state powers"

- *Invited talk*, Electronics & Telecommunications Research Institute, Dec. 2024

- *Invited talk*, SNU Quantum Information Theory Seminar, Dec. 2024*
- *Invited talk*, IBM-Yonsei Qiskit Fall Fest, Nov. 2024*
- *Contributed talk*, Annual Meeting of Korean Mathematical Society, Oct. 2024
- *Poster*, QIP 2025, Feb. 2025

"Mutual information maximizing quantum generative adversarial network"

- *Invited talk*, Triangle Quantum Computing Seminar, NC State University Quantum Initiative, Nov. 2023*

"Estimating quantum mutual information through a quantum neural network"

- *Invited talk*, CS Katha Barta, National Institute of Science Education and Research Bhubaneswar, Aug. 2023*

"Quantum Rényi entropy functionals for bosonic Gaussian systems"

- *Poster*, QIP 2022, Mar. 2022

"High-dimensional private quantum channels and regular polytopes"

- *Invited talk*, KISTI-KU-SNU Joint Workshop, Sep. 2023*
- *Invited talk*, SNU Quantum Information Theory Seminar, Aug. 2021*
- *Contributed talk*, Winter Meeting of the Optical Society of Korea, Feb. 2022
- *Contributed talk*, Fall Meeting of the Korean Physical Society, Oct. 2021*
- *Poster*, QIP 2022, Mar. 2022

Tutorials and Public Lectures

"Learning theory in ∞-dimensional quantum systems"

- *Invited talk*, Team QST Summer Workshop, Seoul National University, Aug. 2025

"Introduction to quantum machine learning"

- *Invited talk*, AWS Korea Healthcare & Research Team Seminar, Mar. 2025

"Topics in theoretical quantum computer science"

- *Invited talk*, Shinil High School, Aug. 2024

"Quantum machine learning models for drug library generation"

- *Invited talk*, Yonsei Quantum Computing and Monte Carlo Workshop, Aug. 2024

"$\mathsf{QMA} \stackrel{?}{=} \mathsf{NP}$: The NLTS theorem and the quantum PCP conjecture"

- *Invited talk*, SNU Center for Quantum Network's Channel Capacity Summer Workshop, Jul. 2024

"Minimal data may be sufficient for quantum artificial intelligence"

- *Invited talk*, SNU Department of Mathematical Sciences Seminar, Jun. 2023*

Skills and Technical Experience

Programming Languages: Proficient in C, C++ (Informatics Olympiad), and Python; experienced with Java.

Quantum Software: Proficient in PennyLane and IBM Qiskit (certified); experienced with $Q^{\#}$ and PyZX.

- *IBM Certified Associate Developer*, Quantum Computation using Qiskit, 2023
- *Advanced Achievement*, IBM Quantum Spring Challenge, 2023
- *Advanced Achievement*, Xanadu QHack Coding Challenges, 2023